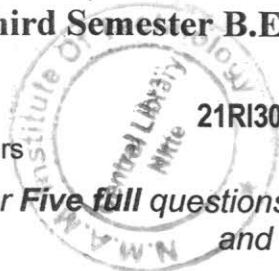


NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Third Semester B.E. (RI) (Credit System) Degree Examinations
April 2023



21RI303 – ANALOG AND DIGITAL CIRCUITS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a neat diagram explain CMOS current source amplifier.	6	L*2	1	1
b)	Define the following operational amplifier characteristics. (i) Offset voltage and current (ii) Circuit bandwidth (iii) Slew rate	8	L1	2	1
c)	Explain the operation of Instrumentation Amplifier with a neat diagram.	6	L2	2	1
2. a)	Explain the working of an OP AMP peak detector circuit with waveforms and mention its applications.	10	L2	2	1
b)	Explain the working of Full wave Precision rectifier with neat circuit diagram and waveforms. Mention its applications.	10	L2	2	1
3. a)	With hysteresis characteristics explain the working of Schmitt trigger circuit. (Inverting)	8	L2	2	1
b)	Explain the working of R-2R ladder D to A converter and derive the expression for V_0 .	6	L2	2	1
c)	With a neat circuit diagram and waveform explain the working of monostable multivibrator using 555 timer.	6	L2	2	1
Unit – II					
4. a)	Convert to hexadecimal and then to binary. (i) 757.25 ₍₁₀₎ (ii) 123.17 ₍₁₀₎	8	L3	3	1
b)	Simplify the following expressions by applying one of the theorems or laws. State the theorem or law used. (i) $F = AB + \bar{A}C + BC$ (ii) $F = A + B(AC + (B + \bar{C})D)$ (iii) $XY + \bar{X}Z + YZ$	12	L3	3	1
5. a)	Find the minimum SOP and minimum POS expressions for the following functions using K map. $f(A, B, C, D) = \sum m(1, 3, 4, 11) + \sum d(2, 7, 8, 12, 14, 15)$	6	L3	3	1
b)	Minimize the following function for POS using K map and realize it using basic gates. $f(A, B, C, D) = \prod m(0, 1, 6, 8, 11, 12) + d(3, 7, 4, 15)$	6	L3	3	1
c)	Simplify the following function using K map and obtain simplified Boolean expressions. (i) $f_1(a, b, c, d) = \sum m(1, 3, 4, 5, 7, 10, 12)$ (ii) $f_2(a, b, c, d) = \sum m(5, 8, 9, 10, 11, 12, 13, 14, 15)$	8	L3	3	1

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|----|----|--|---|----|---|---|
| 6. | a) | Explain Full Adder circuit with truth table and circuit diagram. | 6 | L2 | 4 | 1 |
| | b) | Implement the following functions using 3:8 decoder.
$f_1(a,b,c) = \sum m(0,4,6,7)$
$f_2(a,b,c) = \sum m(1,4,5)$ | 6 | L3 | 4 | 1 |
| | c) | What is multiplexer? Implement the following function using 8:1 MUX.
$f(A,B,C,D) = \sum m(1,2,5,6,9,12)$ | 8 | L3 | 4 | 1 |

Unit – III

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|----|----|--|----|----|---|---|
| 7. | a) | Define Power Amplifier. Explain the types of Power Amplifier. Also write a block diagram of Power Amplifier system. | 10 | L2 | 5 | 1 |
| | b) | Explain the operation of push - pull amplifier with circuit diagram and output waveforms. Derive the power efficiency. | 10 | L2 | 5 | 1 |
| 8. | a) | Explain PWM generation in Power Amplifiers. Mention the advantages of PWM amplifiers. | 10 | L2 | 5 | 1 |
| | b) | Explain Buck Regulators with neat circuit diagram and waveforms. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
